

28/09/26

Signal Processing

• Recap: Modulation ✓

• Example: $x_{\text{mod}}[n] = \cos \omega_0 n \cdot x[n]$ ✓

• Convolution, Differentiation in frequency, Parseval's theorem, Modulation or Windowing theorem ✓

• Revisit z-transform

- $x[n] = a^n u[n];$

- $x[n] = -a^n u[-n-1];$

- $x[n] = \left(\frac{1}{2}\right)^n \cdot u[n] + \left(-\frac{1}{3}\right)^n u[n]$

• Modulation: $x_{\text{mod}}[n] = \underbrace{e^{j\omega_0 n}}_{\text{modulation}} \cdot \underbrace{x[n]}_{\text{signal}}$

$$X_{\text{mod}}(e^{j\omega}) = X(e^{j(\omega - \omega_0)})$$

• Amplitude modulation (AM): $x_{\text{AM}}[n] = \cos \omega_0 n \cdot x[n]$
 $= x[n] \cdot \cos \omega_0 n$

Q: How does $X_{\text{AM}}(e^{j\omega})$ relate to $X(e^{j\omega})$?

A: Note that $\cos \omega_0 n = \frac{1}{2} (e^{j\omega_0 n} + e^{-j\omega_0 n})$

i.e., $x_{\text{AM}}[n] = x[n] \cdot \frac{1}{2} (e^{j\omega_0 n} + e^{-j\omega_0 n})$

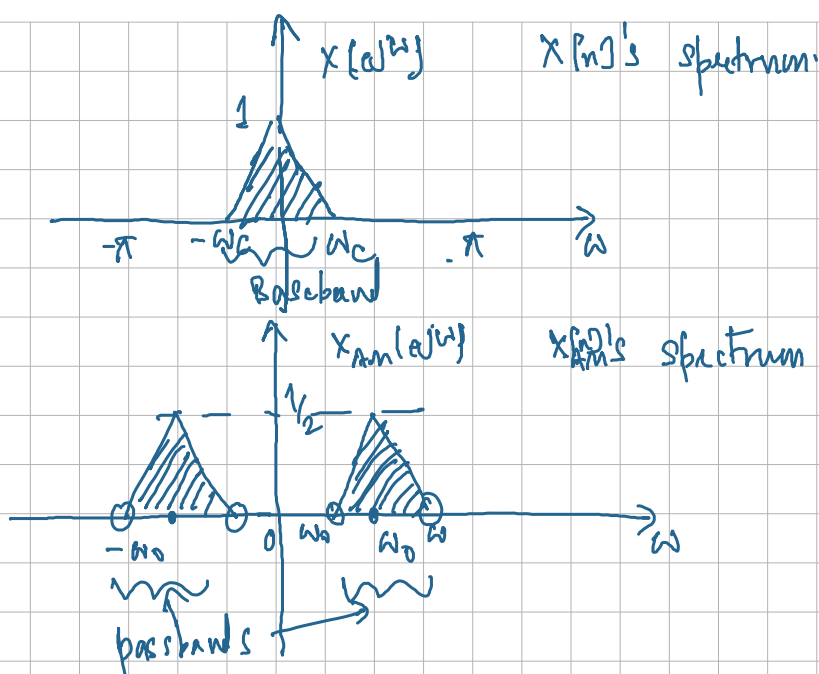
$$\Rightarrow X_{\text{AM}}(e^{j\omega}) = \text{DTFT} \{x_{\text{AM}}[n]\}$$

$$= \text{DTFT} \left\{ \frac{x[n]}{2} e^{j\omega_0 n} + \frac{x[n]}{2} e^{-j\omega_0 n} \right\}$$

$$= \frac{1}{2} (X(e^{j(\omega - \omega_0)}) + X(e^{j(\omega + \omega_0)}))$$

Announcement

- PPR due on 28/09 @ 11:59 pm
- 2pm to 3pm today at EE605



Notes Key to a lot of common systems.

• If $x[n] \leftrightarrow X(e^{j\omega})$, $h[n] \leftrightarrow H(e^{j\omega})$ then
 $y[n] = x[n] * h[n] \leftrightarrow X(e^{j\omega}) \cdot H(e^{j\omega})$

$$Y(e^{j\omega}) = H(e^{j\omega}) \cdot X(e^{j\omega})$$

$$= |H(e^{j\omega})| \cdot \exp(\angle H(e^{j\omega})) \cdot$$

$$|X(e^{j\omega})| \exp(\angle X(e^{j\omega}))$$

$$= |H(e^{j\omega})| \cdot |X(e^{j\omega})| \cdot \exp(\angle H(e^{j\omega}) + \angle X(e^{j\omega}))$$

Note: Both magnitude and phase response impact the "shape" of $Y(e^{j\omega})$.

• Systems with a linear phase response have "useful properties."

$$x[n - n_d] \leftrightarrow e^{-jn_d\omega} X(e^{j\omega})$$

phase response of a system: $H(e^{j\omega}) = 1 \cdot \exp(-jn_d\omega)$

• Differentiation in the frequency domain:

$$x[n] \leftrightarrow X(e^{j\omega})$$

$$n x[n] \leftrightarrow j \frac{dX(e^{j\omega})}{d\omega}$$

$$j \frac{dX(e^{j\omega})}{d\omega} = j \frac{d}{d\omega} \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$= j \sum_{n=-\infty}^{\infty} x[n] \cdot \underbrace{\frac{d}{d\omega} e^{-j\omega n}}_{-jn}$$

$$= j \sum_{n=-\infty}^{\infty} x[n] (-jn) e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{\infty} n \cdot x[n] \cdot e^{-j\omega n}$$

$$= \text{DTFT} \{ n \cdot x[n] \}$$

• Parseval's theorem:

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

• Windowing theorem:

$$x[n] \cdot h[n] \leftrightarrow X(e^{j\omega}) \otimes H(e^{j\omega})$$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta}) H(e^{j(\omega-\theta)}) d\theta$$

• Revisit the z-transform:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$|z| = \underbrace{r}_{>0} e^{j\omega}$$

Ex: $x[n] = a^n \cdot u[n]$

$$\Rightarrow X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} = \sum_{n=-\infty}^{\infty} a^n \cdot u[n] \cdot z^{-n}$$

$$= \sum_{n=0}^{\infty} a^n \cdot z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1 - az^{-1}}; |az^{-1}| < 1$$

$$\Rightarrow \left| \frac{a}{z} \right| < 1 \text{ or } |z| > |a|$$

i.e., ROC is a region outside the circle with radius a

